

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour
Total Marks: 50

Annexure A

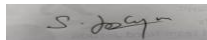
Error Analysis Task

- A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I logesh S 192525023 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Neural Network Performance Analysis Using Mean Squared Error (MSE)

Given

Training MSE = 0.85

Testing MSE = 5.20

Mean Squared Error (MSE) measures the average squared difference between the actual and predicted values. A lower MSE indicates better model performance.

The training MSE (0.85) is much lower than the testing MSE (5.20). This significant difference indicates that the neural network performs very well on the training data but poorly on unseen test data. This situation is commonly known as overfitting.

1. Understanding the Difference Between Training and Testing MSE

Training MSE

Training MSE is the error calculated using the same dataset that was used to train the neural network. A low training MSE means the model has learned the training data effectively.

Testing MSE

Testing MSE is the error calculated on new, unseen data. It measures how well the trained model generalizes to real-world data.

Interpretation

Since

Training MSE = 0.85

and

Testing MSE = 5.20

the testing error is much larger than the training error. This indicates that the model has learned the training data too closely, including its noise and unnecessary details, resulting in poor performance on new data.

2. Possible Reasons for the Difference

A. Overfitting

Overfitting is the most common reason for such a large difference between training and testing errors.

In overfitting:

The model memorizes the training data instead of learning general patterns.

It performs extremely well during training.

It fails to make accurate predictions on unseen data.

Characteristics

Low training error

High testing error

Poor generalization ability

B. Small Training Dataset

If the training dataset is too small:

The model does not see enough examples.

It cannot learn all possible patterns.

It memorizes the available data instead of learning general rules.

As a result, testing performance decreases.

C. Complex Neural Network Architecture

A network with:

Too many hidden layers

Large number of neurons

Excessive parameters

can easily memorize the training data.

Although training accuracy becomes very high, testing performance suffers.

D. Noise in the Training Data

Training data may contain:

Incorrect values

Missing values

Outliers

Measurement errors

The neural network may learn these noisy patterns, reducing its ability to predict new data correctly.

E. Lack of Regularization

Regularization techniques prevent the model from becoming overly complex.

If regularization is not used:

Weights become too large.
The model memorizes training data.
Overfitting occurs.

F. Poor Data Preprocessing

If data is not properly prepared:

Different feature scales
Missing values
Duplicate records
Incorrect encoding
the neural network may not learn efficiently.

G. Improper Train-Test Split

If training and testing datasets have different distributions:

Training data may be easier.
Testing data may contain unseen patterns.
This increases testing error.

H. Excessive Number of Training Epochs

Training for too many epochs causes:

Continuous reduction in training error.
Increase in testing error.
The model begins to memorize the training data.

I. Hyperparameter Selection

Incorrect hyperparameters can affect performance.

Examples include:

Learning rate

Batch size

Number of hidden layers

Activation functions

Optimizer choice

Poor hyperparameter selection may result in poor generalization.

3. Corrective Measures

A. Apply Regularization

Regularization reduces model complexity.

Common methods include:

L1 Regularization

L2 Regularization

Weight Decay

Benefits:

Prevents large weights.

Reduces overfitting.

Improves testing accuracy.

B. Use Dropout

Dropout randomly disables some neurons during training.

Advantages:

Prevents neuron dependency.

Improves generalization.

Reduces overfitting.

C. Early Stopping

Monitor validation error during training.

Stop training when validation error begins increasing.

Benefits:

Prevents over-training.

Saves computation time.

Produces a better model.

D. Increase Training Data

More data allows the model to learn a wider variety of patterns.

Methods include:

Collect additional data.

Data augmentation (where applicable).

Benefits:

Better generalization.

Reduced overfitting.

E. Simplify the Neural Network

Reduce:

Hidden layers

Number of neurons

Trainable parameters

A simpler model is less likely to memorize training data.

F. Improve Data Quality

Perform data preprocessing:

Remove duplicate records.

Handle missing values.

Remove outliers.

Normalize or standardize features.

Correct incorrect values.

High-quality data leads to better predictions.

G. Cross-Validation

Use techniques such as k-fold cross-validation.

Benefits:

Better estimation of model performance.

More reliable evaluation.

Improved hyperparameter tuning.

H. Hyperparameter Tuning

Optimize parameters such as:

Learning rate

Batch size

Number of epochs

Activation functions

Optimizer

Methods include:

Grid Search

Random Search

Bayesian Optimization

I. Feature Selection

Remove irrelevant or redundant input features.

Benefits:

Reduces complexity.

Faster training.

Better prediction accuracy.

4. Expected Outcome After Applying Corrective Measures

After implementing the above improvements:

Training and testing MSE become closer.

Generalization improves.

Prediction accuracy increases.

Overfitting is minimized.

The neural network performs better on unseen data.